MANUSHREE SHAH

AI ENGINEER | FORWARD DEPLOYED AI (FDE) | GENAI & ENTERPRISE AI SOLUTIONS

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PROFESSIONAL SUMMARY

AI Engineer and client-facing AI solution builder with experience architecting, developing, and deploying enterprise-grade Generative AI, Agentic AI, and Machine Learning applications in **fast-paced, 0-to-1 startup environments**. Experienced in working directly with business stakeholders and cross-functional teams to identify high-impact use cases, translate business requirements into technical architectures, and drive end-to-end AI solution delivery from discovery and prototyping to production deployment. Hands-on expertise in LLMs, RAG, multi-agent orchestration, multimodal AI, model fine-tuning, cloud-native application development, and LLMOps. Proven ability to bridge business and engineering teams, take ownership of technical execution, and deliver measurable outcomes through production AI systems across financial services. **Gold Medalist** in Artificial Intelligence, with postgraduate specialization in Cloud AI Engineering & AIOps from IIT Roorkee.

CORE COMPETENCIES

- **Forward Deployed AI Engineering:** Client Discovery, Business Requirement Analysis, Use Case Identification, Solution Architecture, Rapid Prototyping, End-to-End Implementation, Production Deployment, Technical Stakeholder Management.
- **Generative & Agentic AI:** LLM Applications, RAG, Agentic Workflows, Multi-Agent Orchestration, Tool Calling, Prompt Engineering, Context Engineering, Multimodal AI, SLM Fine-Tuning.
- **Enterprise AI Solutions:** AI-Powered Process Automation, Intelligent Document Processing, Knowledge Systems, Compliance Automation, Voice AI, Conversational AI, Decision Support Systems.
- **AI Engineering & Operations:** LLMOps, Model Evaluation, Guardrails, Performance Optimization, Monitoring, CI/CD, Scalable APIs, Production Reliability.

TECHNICAL SKILLS

- **Programming & Machine Learning:** Python, SQL, PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, TabNet, Ensemble Learning, Predictive Modeling, Computer Vision (YOLO).
- **Generative AI & Agentic Systems:** LangChain, LangGraph, n8n, Gemini APIs, LLM Integration, RAG, Agentic Workflows, Tool Calling, Prompt Engineering (CoT, Few-Shot), QLoRA Fine-Tuning, Gemma 3, LLaMA, Multimodal AI.
- **Retrieval & Data Systems:** Pinecone, FAISS, PostgreSQL, Vector Databases, Embeddings, Data Processing, ETL Pipelines.
- **Cloud & Backend Engineering:** Google Cloud Platform (Compute Engine, Cloud Run), Flask, REST APIs, NGINX, Docker, VM Provisioning, Microservices, GitHub Actions, CI/CD.
- **AI Operations:** LLM Evaluation, Model Performance Monitoring, Latency Optimization, Model Versioning, Guardrails, Deployment and Production Monitoring.

PROFESSIONAL EXPERIENCE

Investigen.ai | AI Product Engineer
July 2025 – Present

Own technical delivery of enterprise AI solutions for financial services clients, working across business problem discovery, solution architecture, engineering, stakeholder collaboration, and production deployment.

- **Client-Facing AI Solution Architecture:** Collaborate with enterprise stakeholders across Product, Compliance, Marketing, and business teams to identify automation opportunities, translate business challenges into AI use cases, define solution architectures, and drive implementation through production.
- **Agentic AI & Enterprise Automation:** Architected and deployed multi-agent workflows using LangGraph, LangChain, and Gemini APIs, integrating custom tools to automate complex, multi-step financial calculations and planning workflows. Developed Flask-based backend services and supported high-concurrency workloads using NGINX on GCP.
- **Multimodal AI & Intelligent Compliance:** Designed and deployed an end-to-end multimodal processing engine integrating speech-to-text (Whisper), computer vision, and LLM-based evaluation to analyze advertising creatives across text, audio, video, and documents against 40+ SEBI compliance rules, reducing manual review effort by 70%.

- **Voice AI & Enterprise Enablement:** Built an end-to-end AI-powered sales coaching platform integrating STT/TTS, Gemini LLMs, and Pinecone vector retrieval to evaluate sales conversations and deliver actionable, real-time feedback. Improved pitch consistency by approximately 30% across 100+ sales professionals at a major Asset Management Company.
- **LLM Fine-Tuning & LLMOps:** Fine-tuned domain-specific Small Language Models (Gemma 3 4B) using QLoRA for financial classification tasks. Established model evaluation and monitoring pipelines to assess response quality, latency, and performance.
- **Production Engineering & Deployment:** Developed and integrated cloud-hosted AI services, REST APIs, retrieval pipelines, and backend components, focusing on scalability, performance, reliability, and enterprise integration.

Investigen.ai | AI Developer Intern
January 2025 – June 2025

- **RAG & Enterprise Knowledge Systems:** Designed and developed secure Retrieval-Augmented Generation pipelines using Python, Gemini, Pinecone, and FAISS to retrieve and synthesize information from complex financial regulations and domain-specific knowledge sources.
- **AI-Powered Business Automation:** Developed and shipped an automated investment proposal generation engine using Python and Gemini APIs, standardizing client deliverables and reducing manual document generation effort.
- **End-to-End AI Application Development:** Integrated LLM APIs, retrieval systems, and backend services to convert business requirements into functional AI-powered applications.

Bandhan AMC | Product Intern
June 2024 – October 2024

- **Data & Workflow Automation:** Developed Python-based ETL pipelines to automate daily operational data processing and reporting workflows.
- **Business Analytics:** Built data visualization workflows to analyze sales performance metrics and surface actionable insights for business stakeholders.

EDUCATION

- **Postgraduate Certification in AI Engineering on Cloud & AIOps**
  Indian Institute of Technology (IIT), Roorkee | September 2025 – June 2026
  Focus: Cloud-native AI architecture, MLOps/LLMOps, scalable microservices, containerization, CI/CD, model deployment, and monitoring.
- **B.Sc. (Hons.) in Artificial Intelligence**
  SVKM's NMIMS University, Navi Mumbai | August 2021 – June 2025
  Gold Medalist & Best Student Awardee | CGPA: 3.95/4.0

RESEARCH & PUBLICATIONS

- **Evaluating Predictive Maintenance with Multiclass Categorical Data Constraints** — Springer, IISU ASSET 2025 .
  Developed deep learning and ensemble predictive maintenance models (TabNet, SMOTE) for rail systems, benchmarking failure-prediction accuracy on sparse categorical-multiclass datasets. (PyTorch)
- **AI-Enabled Automated Inspection of Reinforced Concrete Cement (RCC)** — IEEE Xplore, ICSPCRE-2024, NIT Rourkela
  Automated rebar alignment inspection using object detection and computer vision (YOLO, Roboflow, Weights & Biases).
- **Automating DataManipulationTasks inMS Excel: Comparative Analysis of Gemini and Llama LLMs** — IEEE Xplore, CISCT-2024
  Benchmarked Gemini and Llama models on few-shot and zero-shot code generation tasks for Excel manipulation, evaluating across multiple NLP metrics.

CERTIFICATIONS & LEADERSHIP

- **Microsoft Certified: Azure AI Fundamentals** | May 2024
- **Student Council President, NMIMS Navi Mumbai** | April 2023 – May 2024 - Led a 50+ member team and organized 15+ AI/DS workshops and hackathons, building engagement around technology and innovation.